On a Fast Implementation of a 2D-Variant of Weyl's Discrepancy Measure

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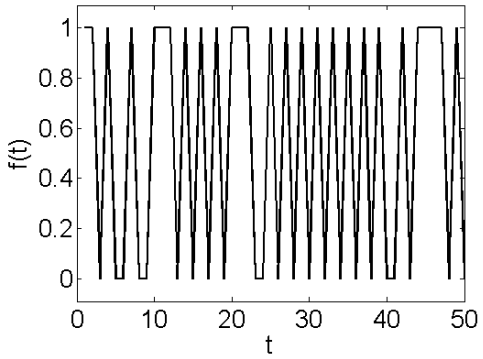
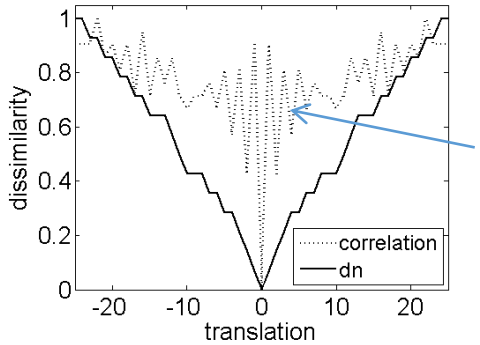
Motivation

- Dissimilarity measure for pattern matching with
  - [C1] zero for identical match
  - [C2] motonotic w.r.t displacements
  - [C3] bounded slope at global minimum (Lipschitz condition)





non-monotonicity

- Non-monotonicity effect for a wide range of standard measures [MSB2011]

[MSB2011] B. Moser, G. Stübl, and J. L. Bouchot. On a Non-Monotonicity Effect of Similarity Measures. In M. Pelillo and E.R. Hancock, editors, 1st Int. Conf. on Similarity-based Pattern Recognition, SIMBAD'11, vol. 7005 of LNCS, pp. 46--60, Springer Berlin Heidelberg, 2011.

Motivation

- Hermann Weyl‘s discrepancy measure [Wey1916] can be extended to similarity measure which satisfies [C1], [C2], [C3] as shown in [Mos2011]
- Applications
  - Stochastics of Random Walks (by this solution of a 65 years old combinatorial problem [Mos2014])
  - Stable metric for Event-Based Sampling (Integrate & Fire etc), [MN2014]
  - robust unimodal MRT registration

[Wey1916] Weyl, H.: Über die Gleichverteilung von Zahlen mod. Eins. Math. Ann. (1916)
[Mos2011] B. Moser. A similarity measure for image and volumetric data based on Hermann Weyl's discrepancy. IEEE Trans. Pattern Analysis & Machine Intelligence, vol. 33, nr. 11, pp. 2321-2329, 2011.
[Mos2014] B. Moser, The Range of a Simple Random Walk on Z: An Elementary Combinatorial Approach, Electronic Journal of Combinatorics, 21(4):#P4.10, 2014
[MN2014] B. Moser and T. Natschläger, On Stability of Distance Measures for Event Sequences Induced by Level-Crossing Sampling, IEEE Trans. on Signal Processing, 62(8):1987--1999, 2014.

Interpretation as Diameter in 1D

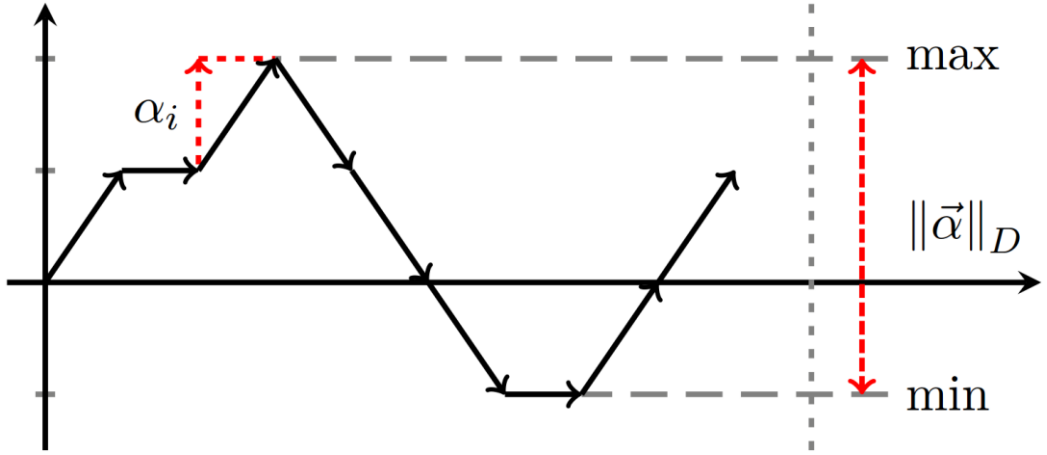
- Dual view in 1-dim based on extremal property
  - maximal partial sum over system of subsets (hypergraph)

$$\|f\|_{\mathcal{C}}^{(n)} = \sup_{C \in \mathcal{C}} \left| \int_C f d\mu \right|$$

hypergraph

  - diameter

$$\|\alpha\|_D = \max_{0 \le k \le n} \sum_{i=0}^{k} \alpha_i - \min_{0 \le k \le n} \sum_{i=0}^{k} \alpha_i$$



Discrepancy Measure, 2D Variant

- Generalization of “diameter concept” to images

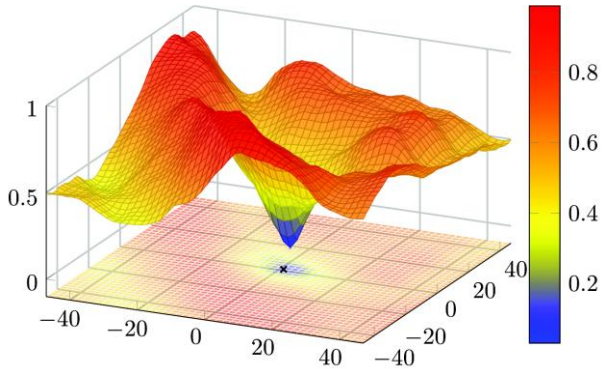
$$
\begin{aligned}
\|f\|_D := \quad \max \quad \Bigg\{ & \max_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(i,j)}\} - \min_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(i,j)}\} \ , \\
& \max_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(N-i,j)}\} - \min_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(N-i,j)}\}, \\
& \max_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(i,M-j)}\} - \min_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(i,M-j)}\}, \\
& \max_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(N-i,M-j)}\} - \min_{0\le k\le N, 0\le l\le M} \{\sum_{i=0}^{k}\sum_{j=0}^{l} I_{(N-i,M-j)}\} \Bigg\}
\end{aligned}
$$

- based on four integral images (components)

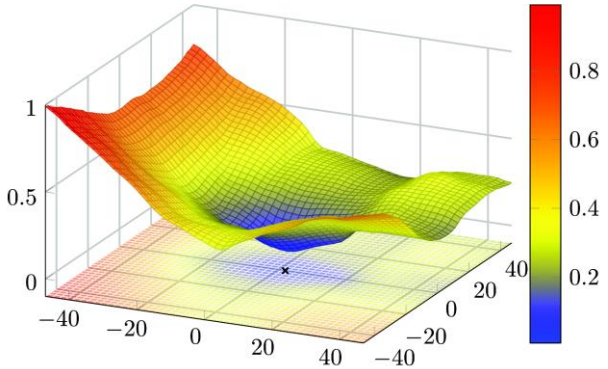
Robustness

- Similarity Map based on Template Matching with sliding window technique [Mot2015]

NCC (norm. cross-correlation)



DN (discrepancy measure)



[Mot2015] C. Motz. Pattern matching based on Hermann Weyl's discrepancy norm. Hagenberg, 2015.

Computational Complexity

- O(number of pixels) for integral image variant

- Goal of reducing computational costs by
  - Restructuring of computation in order to avoid redundancies
  - Exploitation of parallelization techniques

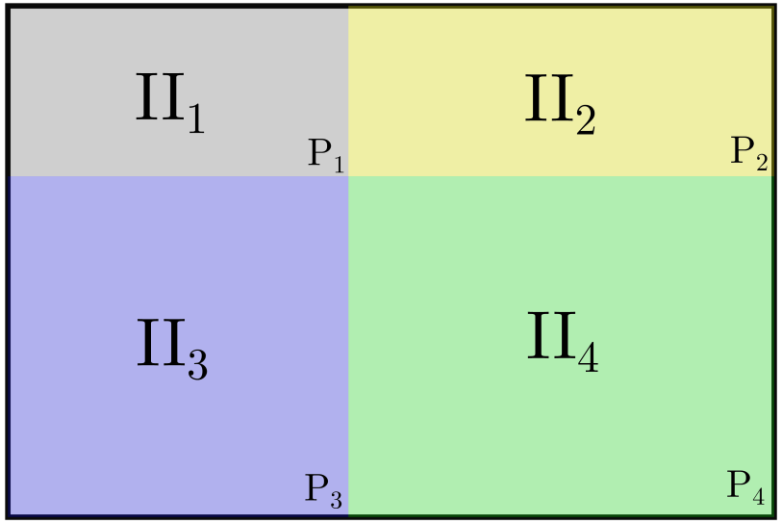
- Result: speedup of 16 [Mot2015]

[Mot2015] C. Motz. Pattern matching based on Hermann Weyl's discrepancy norm. Hagenberg, 2015.

Optimization: Integral Images

- Compute any rectangular area with 4 accesses
- Discrepancy measure 2D: 4 integral images

- Integral images starting from different corners
  - Redundant
  - Relationship through constants



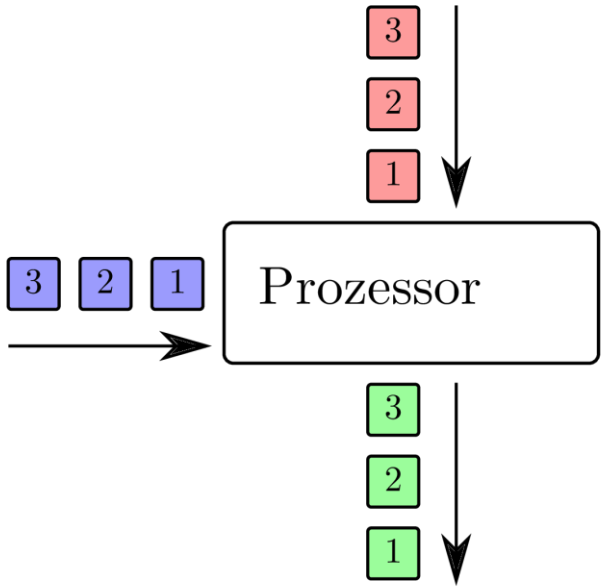
Optimization: Compare Operation

- Minimum / maximum for a fixed row

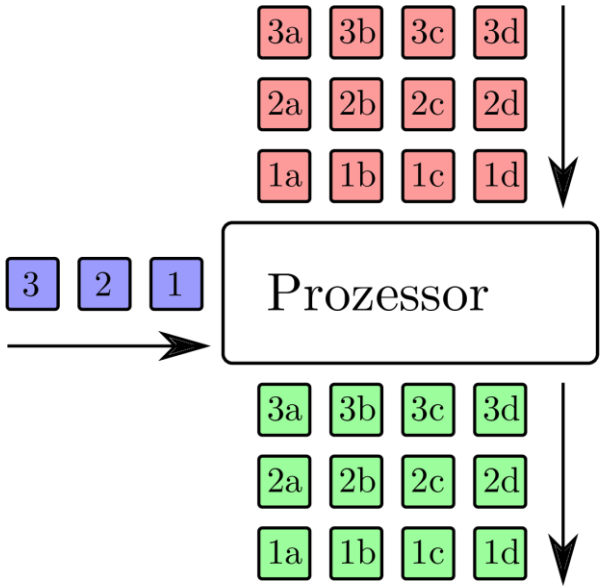
- Two components
  - differ only in constant and sign
  - Exception: one alternative value possible
- Min / Max difference
  - Sign change swap Min / Max
  - deduction possible with 1 more compare
  - Additional alternative requires 1 sub/add per row/column

Parallelization

single instruction, single data
→ SISD



single instruction, multiple data
→ SIMD



[Fly1972] M. Flynn: Some Computer Organizations and Their Effectiveness, IEEE Trans. Computers, C-21, S. 948–960, 1972.

Vectorization: Analysis

- Parallelization of integral images / summed area tables
  - GPU [BHM2010]: for large data
  - Multi-core [Zha2010]: processor restrictions, vectorization not efficient

- SIMD on x86/x64
  - SSE: 128 bit = 4x 32 bit integer / float
  - AVX: 256 bit = 8x 32 bit integer / float

[BHM2010] B. Bilgic, B. K. P. Horn and I. Masaki. Efficient integral image computation on the GPU. Intelligent Vehicles Symposium (IV), 2010 IEEE, San Diego, 2010, pp. 528-533.
[Zha2010] Nan Zhang: Working towards efficient parallel computing of integral images on multi-core processors. Computer Engineering and Technology (ICCET), Chengdu, 2010, pp. V2-30-V2-34.

Vectorization: Scheme

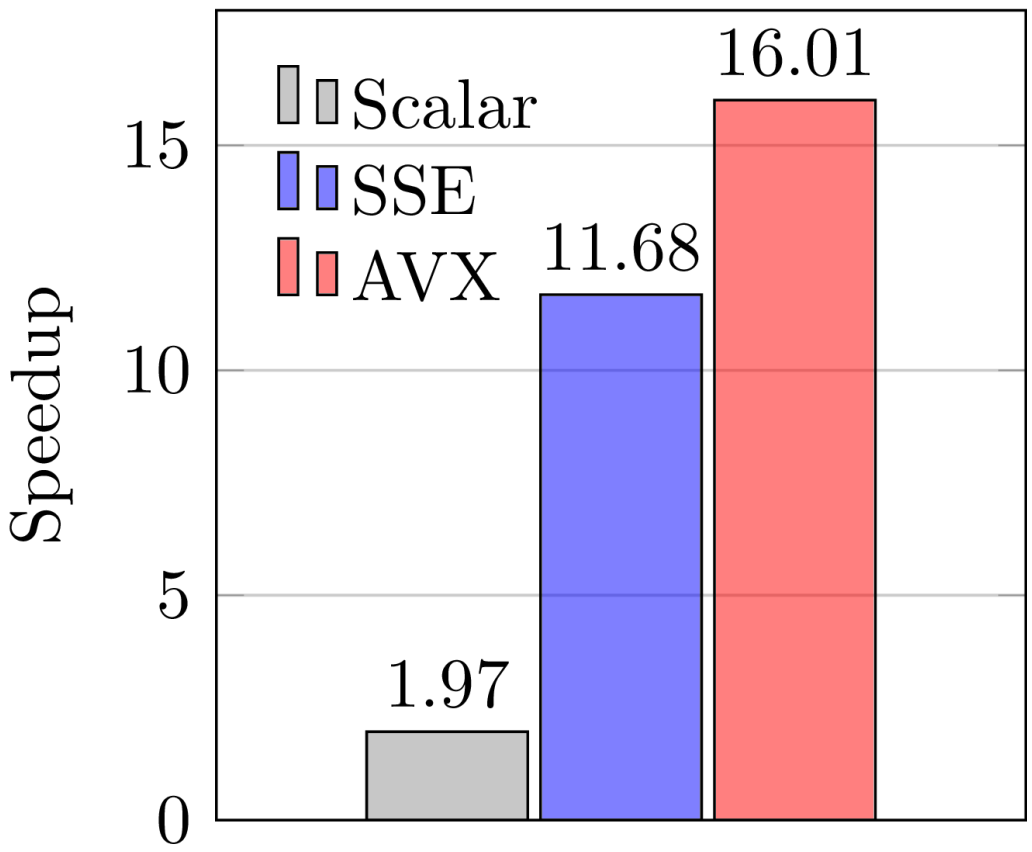
- Dependent on 3 values
- Offset Px + O – PL ideal for vectorization (only 1 broadcast overhead)
- Partial Prefix Sum X1-X4
  - 2 shifts
  - 2 adds
  - log(vector length)

	PL	P1	P2	P3	P4		
	O	X1	X2	X3	X4		

Input	x1	x2	x3	x4
Shift	0	x1	x2	x3
Add	x1	x12	x23	x34
Shift	0	0	x1	x12
Add	x1	x12	x123	x1234

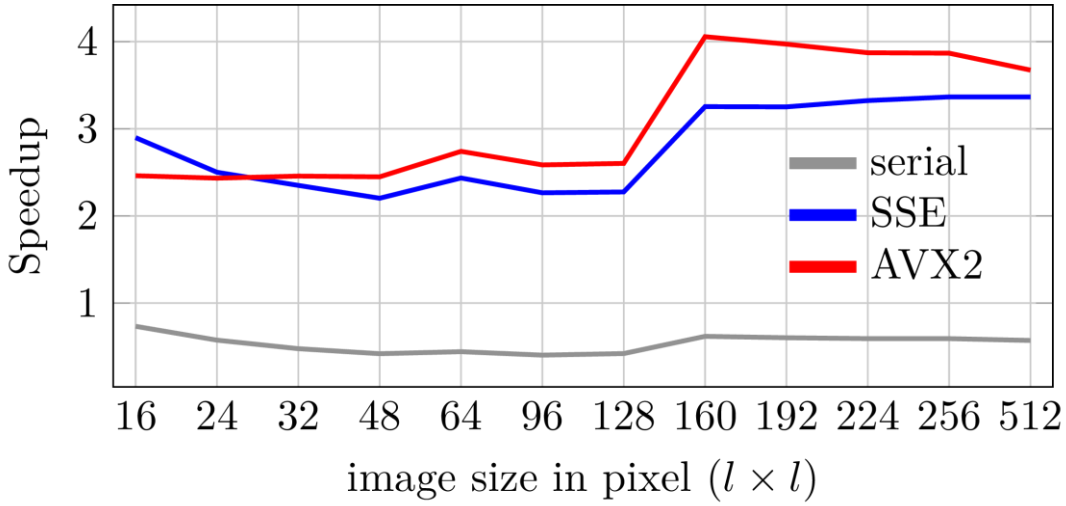
Results: Discrepancy Measure

- Average execution time for calculating similarity map
- Includes vectorization of difference building
- Speedup of 16
  - Algorithmic: 2x
  - Vectorization: 8x
- Theoretical speedup of AVX: 8x → efficient



Results: Integral Images

- Reference: OpenCV
- Speedup of 2.5 to 4
- AVX and SSE similar because of overhead
- Applicable to general concept of Summed Area Tables



Summary

- Optimization steps

Before	After
4 integral images	1 integral image
Min / Max of each integral images	Deduce Min / Max row / column based
4 passes	2 passes (1 possible)

- Results Vectorization

Method	Speedup
Discrepancy Measure	16
Summed Area Tables	2.5 – 4.0

Outlook

- Further algorithmic analysis
  - Relationship between components
- Vectorization
  - Embedded Systems: ARM architecture
  - Other vectorization schemes?
- Template matching
  - Combine with approximation methods

Questions

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